

Revisão de Figuras do Manuscrito

IEEE Access — *Context-oriented Synthesis of Salt Domes*

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Data de revisão: 15 de setembro de 2026

Tabela-Resumo

Status de cada figura

Fig.	Arquivo	Uso no manuscrito	Status	Ação?
1	terre1.png	Fig. 1 — Pipeline	OK	
2	terre2.png	Fig. 2 — VAE Masks	OK	
3	terre3.png	Fig. 3 — Edge/Boundary	OK	
4	terre4.png	Fig. 4 — Border zones	OK	
5	terre5.jpg	Fig. 5 — Texture synthesis	OK	
6	terre6.png	Fig. 6 — Context zones	OK	
7	terre7.png	Nao usada no _v7.tex atual	NAO USADA	Verificar
8	terre8.png	Sem referencia encontrada	NAO USADA	Verificar
9	terre9.png	Synthetic Samples	OK	
10	terre10.png	Quantitative analysis	OK	
11	terre11.png	Sem referencia encontrada	NAO USADA	Verificar
B1	terre.jpg	Foto — Luciano D. Terres	OK	
B2	schar.jpg	Foto — Jacob Scharcanski	OK	

Legenda de cores:

OK

Atencao

Problema

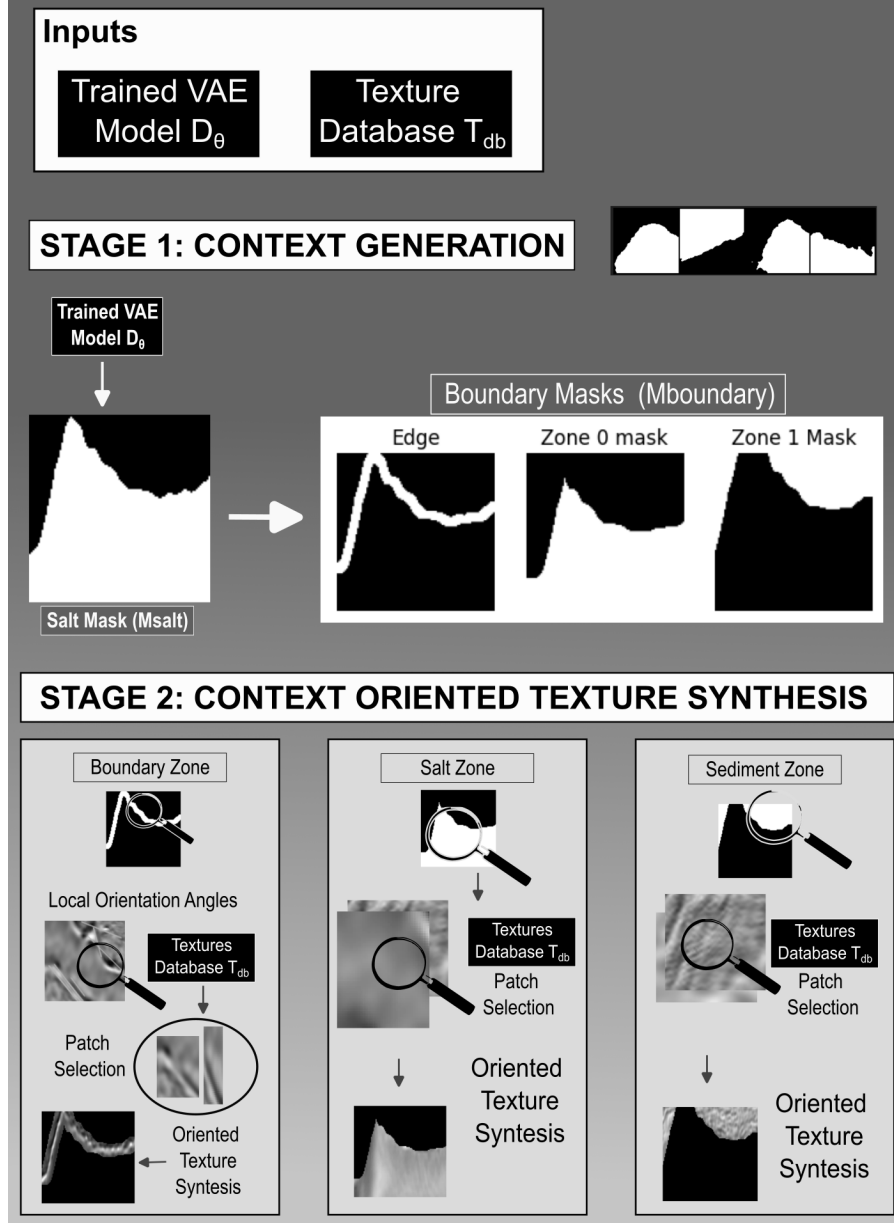
Nao usada

Revisao Individual das Figuras

Figura 1 `terre1.png`

Label: `fig:synthesis_pipeline`

Status: OK



Legenda: “Complete pipeline of the proposed context-oriented seismic image synthesis methodology, showing the integration of VAE-based mask generation and the context-aware texture synthesis method.”

Citacao no manuscrito (`_v7.tex`, 1.109):

“[...] The proposed method focuses on context zones and on the characteristics of seismic images with salt domes, ensuring that the synthetic samples preserve the structural and textural properties of real seismic imaging data. **Figure 1 illustrates the complete synthesis pipeline**, showing how the VAE-generated masks are combined with context-oriented texture synthesis to produce realistic synthetic seismic images.”

Checklist IEEE Access:

- ☐ Resolucao ≥ 300 DPI (fotografias) / ≥ 600 DPI (line art)
- ☐ Sem fundo cinza / sem artefatos de captura de tela
- ☐ Texto legivel apos reducao para coluna simples ($\approx 8,5$ cm)
- ☐ Exportado diretamente do programa-fonte (nao screenshot)
- ☐ Cores funcionam em escala de cinza

Notas de revisao:

Figura 2 terre2.png

Label: fig:vae1

Status: OK



Legenda: “Samples of generated structural salt masks. The Variational Autoencoder learns the features distribution and produces new structural masks.”

Citacao no manuscrito (_v7.tex, l. 121 e l. 131):

- (1) “The proposed data augmentation scheme detects and generates new rock geometries with a VAE, as illustrated in **Fig. 2**.”
- (2) “**Fig. 2** illustrates a dataset of seismic images used to generate new salt dome masks. The natural intersection between two different rocks are their boundaries, and in the case of salt rocks, this boundary usually is apparent, as Fig. 3 shows.”

Checklist IEEE Access:

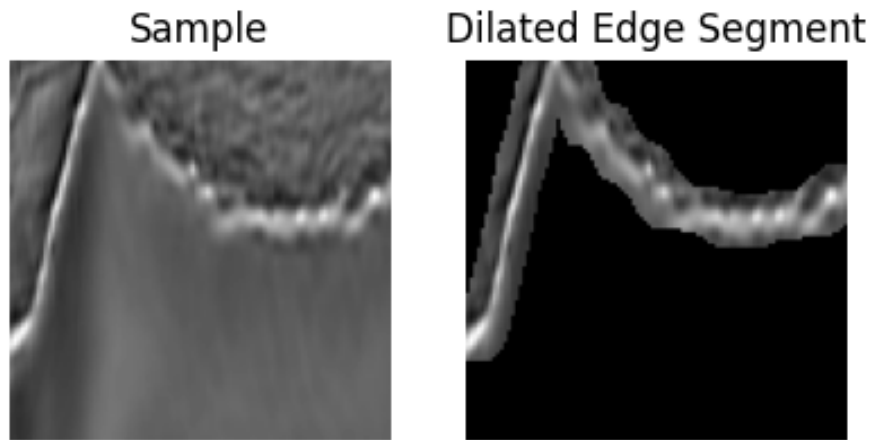
- ☐ Resolucao ≥ 300 DPI
- ☐ Sem fundo cinza / sem artefatos
- ☐ Texto legivel na largura de coluna
- ☐ Exportado diretamente do programa-fonte
- ☐ Cores funcionam em escala de cinza

Notas de revisao:

Figura 3 terre3.png

Label: fig:edge1

Status: OK



Legenda: “Sample of the boundaries identification between salt and other sediments represented in a dilated edge segment.”

Citacao no manuscrito (_v7.tex, 1.131 e 1.280):

- (1) “[...] in the case of salt rocks, this boundary usually is apparent, **as Fig. 3 shows**. During the VAE training process, we use only salt masks with a considerable amount of salt, varying from 10 to 90 percent of the total area.”
- (2) “[...] a boundary strip of 5 pixels total width. **As Fig. 3 shows**, this strip is an area of high seismic contrast appearing as light and dark bands.”

Checklist IEEE Access:

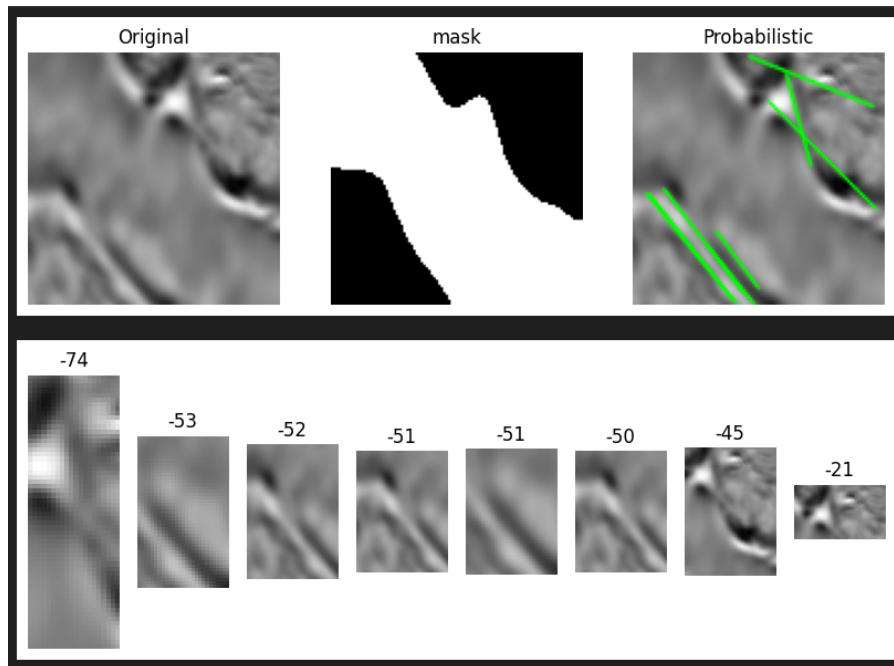
- ☐ Resolucao ≥ 300 DPI
- ☐ Sem fundo cinza / sem artefatos
- ☐ Texto legivel na largura de coluna
- ☐ Exportado diretamente do programa-fonte
- ☐ Cores funcionam em escala de cinza

Notas de revisao:

Figura 4 terre4.png

Label: fig:line1

Status: OK



Legenda: “The green lines at original image identify border zones and angles to guide the patch selection in the patch database construction process.”

Citacao no manuscrito (_v7.tex, 1.280):

“[...] As Fig. 3 shows, this strip is an area of high seismic contrast appearing as light and dark bands. Also, various parallel lines make up this boundary, and **Fig. 4 shows their angles**. We use a dataset composed of edge pieces and their corresponding angles as input for the boundary synthesis process.”

Checklist IEEE Access:

- ☐ Resolucao ≥ 300 DPI
- ☐ Sem fundo cinza / sem artefatos
- ☐ Linhas verdes visiveis e nitidas
- ☐ Exportado diretamente do programa-fonte
- ☐ Cores funcionam em escala de cinza

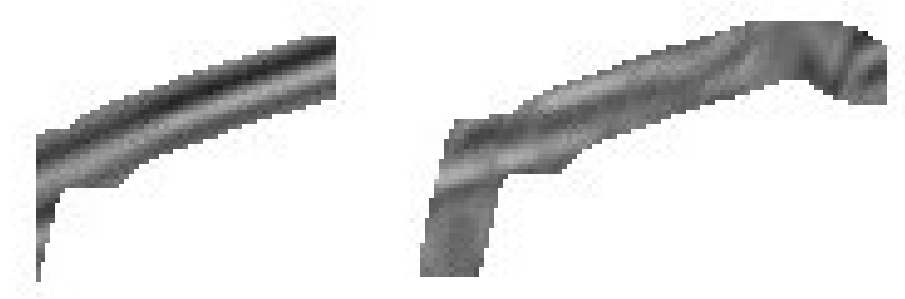
Notas de revisao:



Figura 5 terre5.jpg

Label: fig:edge2

Status: OK



Legenda: “The Construction of new boundaries in the generated seismic images uses texture synthesis based on selected samples from the seismic image patches database.”

Citacao no manuscrito (_v7.tex, 1. 280):

“[...] Then, for each edge segment in the mask of the seismic texture image to be synthesized, we synthesize its texture based on the patch with the most similar angle (see Fig. 5). We repeat this process until all the segments of the edge band are covered.”

Checklist IEEE Access:

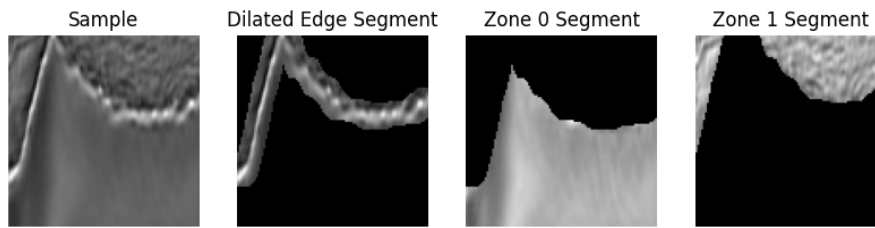
- ☐ Resolucao ≥ 300 DPI (JPEG — verificar compressao)
- ☐ Sem artefatos JPEG visiveis
- ☐ Considerar converter para PNG (sem perdas)
- ☐ Exportado diretamente do programa-fonte
- ☐ Cores funcionam em escala de cinza

Notas de revisao:

Figura 6 terre6.png

Label: fig:source1

Status: OK



Legenda: “Real seismic saline images divided by context zones used as input to the texture synthesis process.”

Citacao no manuscrito (_v7.tex, 1.296):

“[...] We divide the seismic image synthesis into steps to conform to the specific characteristics of this kind of seismic image, such as the edge, saline rock, and conventional rock zones, as **Fig. 6** shows.”

Checklist IEEE Access:

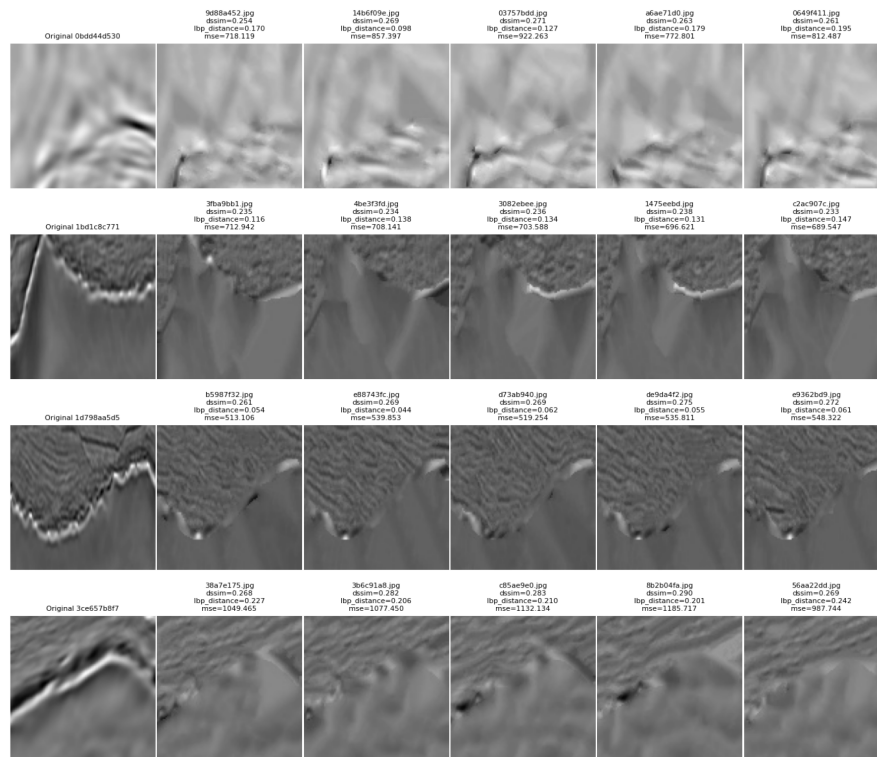
- ☐ Resolucao ≥ 300 DPI
- ☐ Zonas de contexto claramente delimitadas e legiveis
- ☐ Sem fundo cinza / sem artefatos
- ☐ Exportado diretamente do programa-fonte
- ☐ Cores funcionam em escala de cinza

Notas de revisao:

Figura 9 terre9.png

Label: fig:synthetic_samples

Status: OK



Legenda: “Synthetic Seismic Samples.”

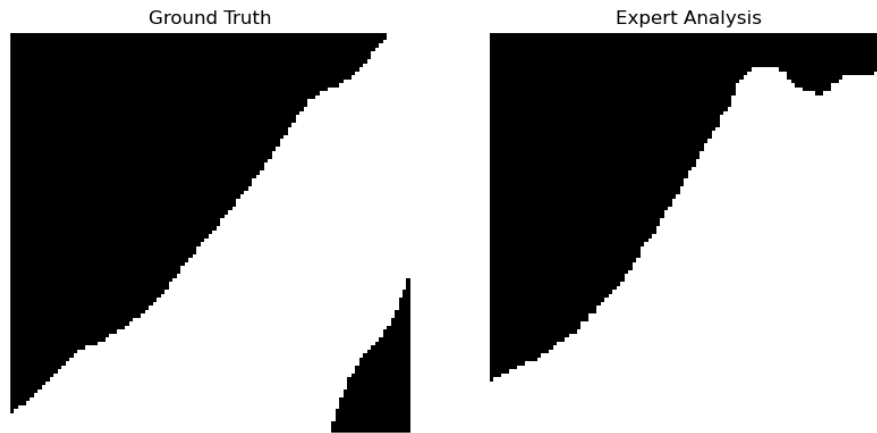
Citacao no manuscrito (_v7.tex, 1.401):

“[...] we evaluate the overlap between the experts selections of pixels corresponding to the salt bodies in synthetic images and synthetic images masks, as illustrated in Fig. 9.”

Checklist IEEE Access:

- ☐ Resolucao ≥ 300 DPI
- ☐ Sem fundo cinza / sem artefatos
- ☐ Amostras sinteticas visualmente representativas
- ☐ Exportado diretamente do programa-fonte
- ☐ Cores funcionam em escala de cinza

Notas de revisao:

Figura 10 terre10.png**Label:** fig:expert2**Status:** ATENCAO — verificar qualidade

Legenda: “Quantitative analysis illustration using sample measurements for comparing the ground truth and expert analysis: Precision: 0.88; Recall: 0.87; F1-score: 0.87.”

Citacao no manuscrito (_v7.tex, 1.401):

“[...] **Fig. 10** illustrates the ground truth image and the specialist evaluation of the salt bodies locations, and the computed precision, recall, and F1-score measures for this particular case.”

Checklist IEEE Access — PRIORIDADE:

- ☐ Resolucao ≥ 600 DPI (contem line art / texto)
- ☐ Exportado diretamente do programa-fonte (**NAO screenshot**)
- ☐ Metricas (0.88 / 0.87) legiveis na impressao
- ☐ Sem fundo cinza
- ☐ Cores funcionam em escala de cinza

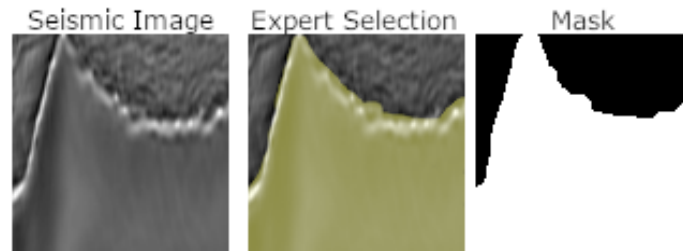
Notas de revisao:

Figuras Nao Referenciadas em _v7.tex

Atencao: As figuras abaixo existem na pasta `images/` mas **nao foram encontradas** em nenhuma `\includegraphics` do arquivo `_v7.tex` atual. Verifique se devem ser incluidas, substituidas ou removidas.

terre7.png — Nao usada no _v7.tex

Nota: esta figura aparece em versoes anteriores do manuscrito (`docs/Responses/r1c2-r2c5/_v7.tex`).



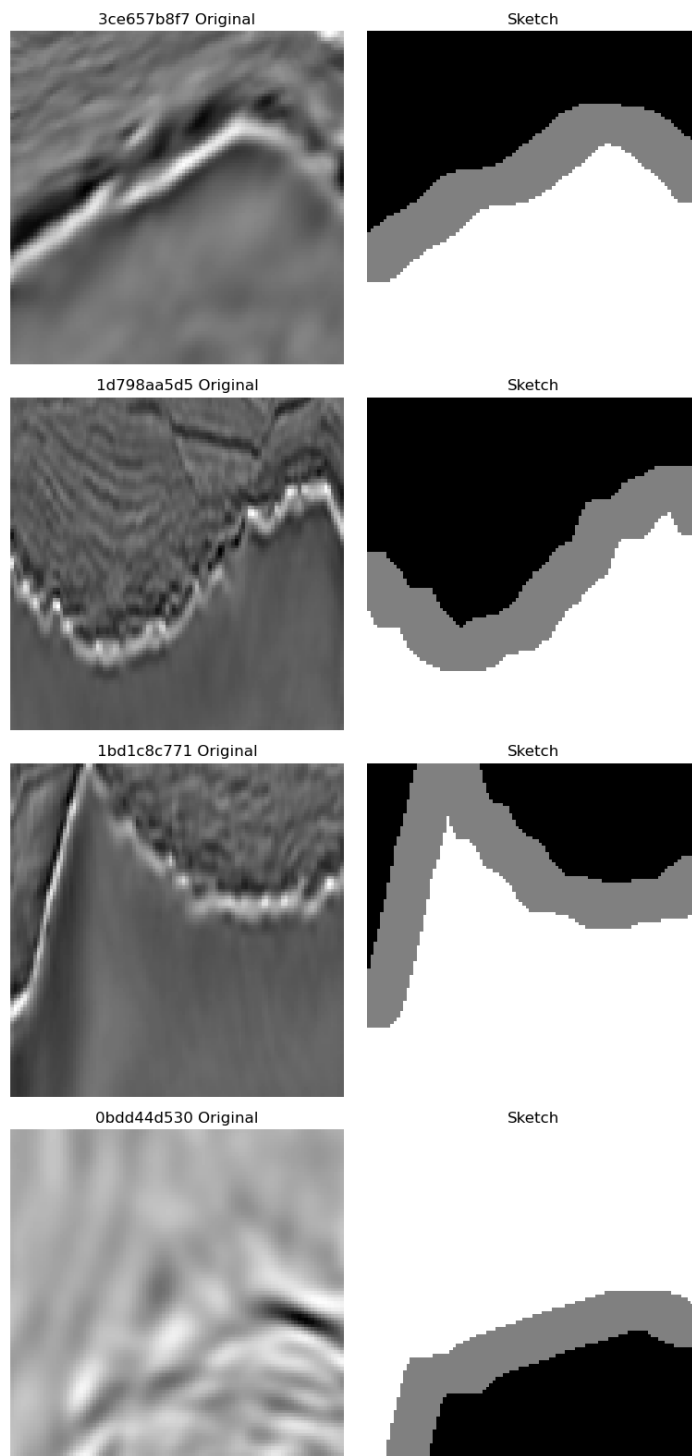
“Comparison between a real seismic image sample, and the geoscience specialist selection in terms of saline area and ground truth (saline mask).”

Decisao:

☐ Adicionar ao manuscrito ☐ Manter removida ☐ Substituir por outra versao

terre8.png — Nao referenciada

Nenhuma referencia encontrada em nenhuma versao do manuscrito. Verificar se e uma versao alternativa de outra figura ou pode ser excluida.



terre8.png — sem referencia encontrada

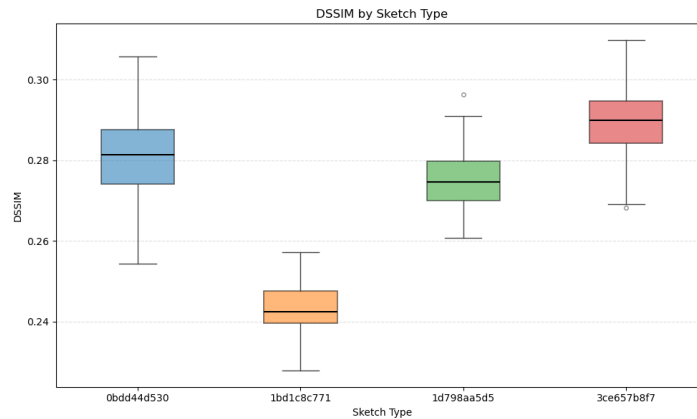
Decisao:

☐ Adicionar ao manuscrito ☐ Excluir arquivo ☐ Arquivar em pasta separada



terre11.png — Nao referenciada

Nenhuma referencia encontrada em nenhuma versao do manuscrito. Verificar se e uma versao alternativa de outra figura ou pode ser excluida.



terre11.png — sem referencia encontrada

Decisao:

☐ Adicionar ao manuscrito ☐ Excluir arquivo ☐ Arquivar em pasta separada

Fotos de Biografia

terre.jpg — Luciano D. Terres



Foto de biografia — Luciano

- ☐ Dimensao: 1in $\times$ 1.25in
- ☐ Qualidade $\geq$ 300 DPI
- ☐ Fundo neutro

schar.jpg — Jacob Scharcanski



Foto de biografia — Jacob

- ☐ Dimensao: 1in $\times$ 1.25in
- ☐ Qualidade $\geq$ 300 DPI
- ☐ Fundo neutro

Resumo de Acoes Necessarias

Prioridade	Arquivo	Acao	Concluido?
Alta	terre10.png	Verificar exportacao direta (≥ 600 DPI); re-exportar se necessario	☐
Alta	terre5.jpg	Considerar conversao para PNG; verificar artefatos JPEG	☐
Media	terre7.png	Decidir inclusao ou exclusao do _v7.tex	☐
Media	terre8.png	Verificar proposito; excluir ou arquivar	☐
Media	terre11.png	Verificar proposito; excluir ou arquivar	☐

Observacoes gerais: